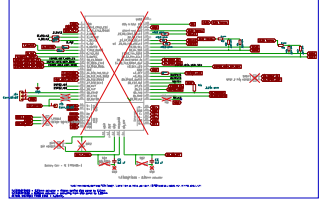
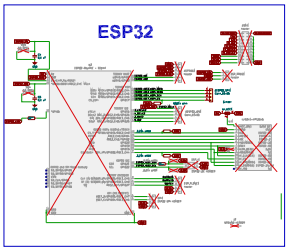
T-DSP Desktop Soundcard

POWER LED

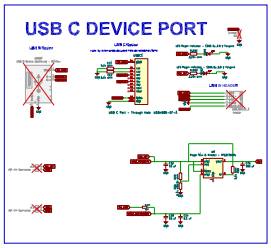
TEENSY 4



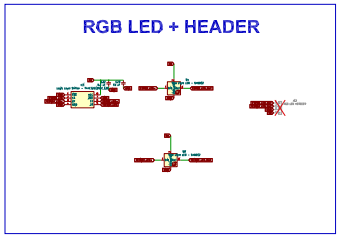
ESP32



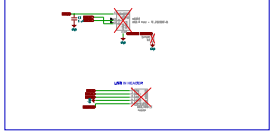
USB C DEVICE PORT



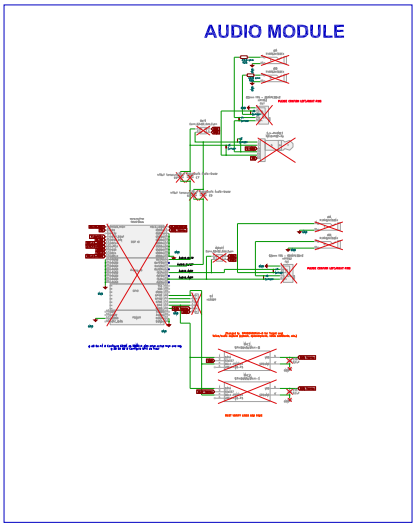
RGB LED + HEADER



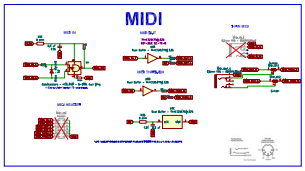
USB A HOST PORT



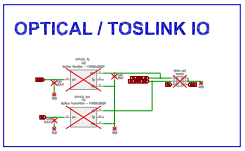
AUDIO MODULE



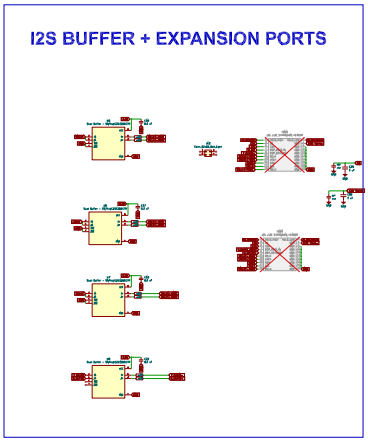
MIDI



OPTICAL / TOSLINK IO

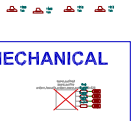


I2S BUFFER + EXPANSION PORTS



SWITCHES

MECHANICAL



HAT MOUNTING POINTS

KiCad 8-Layer Stackup:
F.Cu – TOP: Components, ANALOG routes to jacks, ground fill
In1.Cu – GND: Solid ground plane
In2.Cu – SIG1: TDM/high-speed digital, ground fill
In3.Cu – GND: Solid ground plane (isolation!)
In4.Cu – SIG2: More TDM/digital signals, ground fill
In5.Cu – PWR: Power planes (3.3V/5V)
In6.Cu – GND: Solid ground plane
B.Cu – BOTTOM: Components, ANALOG routes to jacks, ground fill